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Product Validation Report
for
Advanced Visualization Workspace 15 (IntelliSpace Portal 13)

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1 Document Introduction

1.1 Purpose

The purpose of the Validation as called out in the IntelliSpace Portal 13 Validation Plan, was to confirm that the IntelliSpace Portal 13 met its intended use and user needs as described in the User Requirements Specifications (URS) listed in the References section below.
Note: IntelliSpace Portal is rebranded to Advanced Visualization Workspace (AVW) at RfD (Release for limited Delivery), therefore, throughout the document the name "IntelliSpace Portal V13 (ISP13)" is equivalent to the rebranded new name "Advanced Visualization Workspace V15 (AVW15).

1.2 Scope

The objective of the Product Validation Report is to certify that all planned validation activities identified in the Validation Plan DHF380226 have been sufficiently completed to enable the Product to be released once the Manufacturing acceptance process (i.e., successful completion of Acceptance Test, and completion of the Device History Record) is complete.

1.3 References

Reference	Document Title	Document ID
DHF380082	ISP13 - User Requirements Specification	DHF380082
DHF380211	ISP13 PRS	DHF380211
DHF380226	ISP13 Validation Plan	DHF380226
D000943948	ISP13_TRR_Demo_Laptop_Validation	D000943948
D000996308	ISP13_Design Verification Report	D000996308
D000926821	ISP13 PED	D000926821
DHF380228	ISP13 Usability Engineering Plan	DHF380228

1.4 Abbreviations and Definitions

Term	Definition
CCA	Comprehensive Cardiac Analysis
CMR	Cardiac MR
CT	Computed Tomography
DHF	Design History File
DICOM	Digital Imaging and Communications in Medicine
FSE	Field Service Engineer
ICAP	Imaging Clinical Applications and Platforms
ISP	IntelliSpace Portal
CMR	MR Cardiac
MRI	Magnetic Resonance Imaging
NA	Not Applicable
PED	Production Equivalent Device
SMG	Summary map generator
URS	Users Requirements Specification

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2 Deviations from the Validation Plan

VAL1:
CT ASPECTS was validated as described in DHF380226 ISP13 validation plan.
There are no deviation from the validation plan.

VAL2:
CT Liver Analysis (CT Liver), CT CCA, CMR Viewer, CMR Qflow, CMR Spatial and CMR functional Long Axis were validated as described in DHF380226 ISP13 validation plan.
There are no deviation from the validation plan.

VAL3:
CT Calcium Automated Analysis (Auto CS), Spectral AVA, MR Longitudinal Brain Analysis (CVS), CMR Functional LV RV, CMR Temporal and CMR Mapping were validated as described in DHF380226 ISP13 validation plan.
There are no deviation from the validation plan.

2.1 Summary of Variances

VAL1:
No variance

VAL2:
No variance

VAL3:
No variance

2.2 Impact of Variances

VAL1:
No impact

VAL2:
No impact

VAL3:
No impact

3 Validation Results

3.1 Parameters

DHF380226 ISP13 Validation plan

Please Refer to Appendix A for detailed validation results

VAL1	SITE NAME – Internal Validation Only
Validation Start Date:	NOV 2021
Validation End Date:	NOV 2021

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VAL2	SITE NAME – Internal Validation Only
Validation Start Date:	May 2022
Validation End Date:	July 2022

VAL3	SITE NAME – Internal Validation Only
Validation Start Date:	Sept 2022
Validation End Date:	Nov 2022

Evaluation forms and training forms are in section 8.2 and 8.3 of this document.
Defects are logged in Azure DevOps (TFS).

3.1.1 Impact assessment of Incremental Validation

None, there is no impact due to incremental validation.

3.2 Assessment

The following table provides evidence that the validation activities planned in section "Design Validation Top Level Objectives& Statistical Measurement" of the Design Validation Plan for IntelliSpace Portal 13 [DHF380226] have been executed.

3.2.1 Statistical validation

3.2.1.1 *VAL2 and VAL3 – CMR*

Validation Evaluation Forms Data	Ranking							Test
	Perfect	**Good**	**Aacceptable**	**Bad***	**Unacceptable***	**Total No. of Forms (65 required)**	**Pass/Fail of final combined score** Rating of clinically acceptance should be at least 80%.	
CMR Viewer	23	39	3	0	0	**65**	**Passed (By 95.4%)**	1-Proportion test P-Value< 0.05 to reject the Null hypothesis
CMR Qflow	31	30	4	0	0	**65**	**Passed (By 95.4%)**	1-Proportion test

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Validation Evaluation Forms Data	Ranking							Test
	Perfect	**Good**	**Aacceptable**	**Bad***	**Unacceptable***	**Total No. of Forms (65 required)**	**Pass/Fail of final combined score** Rating of clinically acceptance should be at least 80%.	
								P-Value< 0.05 to reject the Null hypothesis
CMR Functional Long Axis	24	37	4	0	0	**65**	**Passed (By 95.4%)**	1-Proportion test P-Value< 0.05 to reject the Null hypothesis
CMR Spatial	10	37	16	2	0	**65**	**Passed (By 90.6%)**	1-Proportion test P-Value< 0.05 to reject the Null hypothesis
CMR Functional LV RV	39	24	2	0	0	**65**	**Passed (By 95.4%)**	1-Proportion test P-Value< 0.05 to reject the Null hypothesis
CMR Mapping	18	41	6	0	0	**65**	**Passed (By 95.4%)**	1-Proportion test P-Value< 0.05 to reject the Null hypothesis
CMR Temporal	12	49	4	0	0	**65**	**Passed (By 95.4%)**	1-Proportion test P-Value< 0.05 to reject the Null hypothesis

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3.2.2 User validation questionnaire

3.2.2.1 *VAL1 - ASPECTS*

The evaluators were asked to grade pass/fail per each step.
See below the analysis for each item. Refer to Appendix A for detailed results of the validation tests.

	Total No. of Forms required - 3	Total No. of cases reviewed per evaluator	Total No. of cases reviewed per evaluator	Pass/Fail	Test
CT ASPECTS	3	3	2	Passed	All steps have passed

Dataset were acquired under the data agreement contract WF-008391.

*All unacceptable clinically (Bad/unacceptable/failed) comments have defects assign to them and are handled by the defect handling process.

3.2.2.2 *VAL2 – CT Liver*

The evaluators were asked to grade pass/fail per each step.
See below the analysis for each item. Refer to Appendix A for detailed results of the validation tests.

	Total No. of Forms required - 3	Total No. of cases reviewed per evaluator	Total No. of cases reviewed per evaluator	Pass/Fail	Test
CT Liver	3	3	2	Passed	All steps have passed

We used the site's own data.

No defect was found during validation.

3.2.2.3 *VAL2 – CT CCA*

The evaluators were asked to grade pass/fail per each step.
See below the analysis for each item. Refer to Appendix A for detailed results of the validation tests.

	Total No. of Forms required - 3	Total No. of cases reviewed per evaluator	Pass/Fail	Test
CT CCA	3	3	Passed	All steps have passed

We used the site's own data.

No defect was found during validation.

3.2.2.1 *VAL3 – CT Auto. CS*

The evaluators were asked to grade pass/fail per each step.
See below the analysis for each item. Refer to Appendix A for detailed results of the validation tests.

	Total No. of Forms required	Total No. of Evaluators	Total No. of cases reviewed per evaluator	Pass/Fail	Test
CT Auto CS	3	3	1	Passed	All steps have passed

We used the site's own data.

No defect was found during validation.

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3.2.2.1 *VAL3 – Spectral AVA*

The evaluators were asked to grade pass/fail per each step
See below the analysis for each item. Refer to Appendix A for detailed results of the validation tests.

	Total No. of Forms required - 3	Total No. of Evaluators	Total No. of cases reviewed per evaluator	Pass/Fail	Test
Spectral AVA	3	3	1	Passed	All steps have passed

We used the site's own data.

No defect was found during validation.

3.2.2.1 *VAL3 – MR Longitudinal Brain Analysis (preliminary named MR CVS)*

The evaluators were asked to grade pass/fail per each step
See below the analysis for each item. Refer to Appendix A for detailed results of the validation tests.

	Total No. of Forms required - 3	Total No. of Evaluators	Total No. of cases reviewed per evaluator	Pass/Fail	Test
MR CVS	3	3	1	Passed	All steps have passed

We used the site's own data.

No defect was found during validation.

3.3 Results Summary

3.3.1 VAL1 - CT ASPECTS

During the validation period a total of three reports of CT ASPECTS were collected.
Each evaluator reviewed two cases: one normal case (no pathology) and one abnormal case previously reported as containing lesions consistent with a diagnosis of both an old and new stroke.
Each evaluator graded a single form according to his impression from the application.
All steps are marked as Pass.

Comments made by the evaluators:

- Dr. Jacobs:
  - I would like annotation of which side is abnormal R vs. L
- Dr. Shreiber:
  - Consider larger zoom for auto. images
  - Consider thinner line for the region segmentation
- Dr. Harsha:
  - Identification of chronic infarcts needs to be worked upon, although technically challenging
  - Identification of suspected acute intracranial hemorrhage should be worked upon as hemorrhage is contra-indication for any stroke revascularization (IV/Mechanical thrombolysis)

Results analysis: Pass
Dissatisfied score: None
Add a description of defect/RFC: see section 4 defects

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3.3.2 VAL2 - CT Liver Analysis

During the validation period a total of three reports of CT liver were collected.
Each evaluator reviewed two cases: one case focusing on diagnosis of liver diseases and second case focusing on liver assessment including surgery resection planning.
Each evaluator graded a single form according to his impression from the application.
All steps are marked as Pass.

No comment was made by the evaluators

Results analysis: Pass
Dissatisfied score: None
Add a description of defect/RFC: None

3.3.3 VAL2 - CT CCA

During the validation period a total of three reports of CT CCA were collected.
Each evaluator reviewed three cases: one case focusing on patients with symptomatic coronary disease, second case focusing on evaluating the patency of coronary artery while planning bypass graft and third case focusing on heart failure disease.
Each evaluator graded a single form according to his impression from the application.
All steps are marked as Pass.

No comment was made by the evaluators.

Results analysis: Pass
Dissatisfied score: None
Add a description of defect/RFC: None

3.3.4 VAL2 - CMR Viewer

During the validation period a total of 65 evaluation forms of general CMR studies were collected, out of which 65 forms were graded as clinically acceptable (Perfect, Good and Acceptable).

The following parameters were the input to the 1-Proportion test (using Minitab 19):
Null hypothesis = there are less than 80% case forms ranked as clinically acceptable.
Reported events = 65
Sample size =65
Confidence level = 95%

Results analysis:
No defect from validation.

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Test and CI for One Proportion

Method

p: event proportion
Exact method is used for this analysis.

Descriptive Statistics

N	Event	Sample p	95% Lower Bound for p
65	65	1.000000	0.954958

Test

Null hypothesis	H_0: p = 0.8
Alternative hypothesis	H_1: p > 0.8

P-Value
0.000

1-Proportion test results: P-value=0.000
P-Value< 0.05 therefore, the Null hypothesis can be rejected. This means that the criterion was met: **at least 80% of the sample was ranked as Clinically Acceptable in a confidence level of 95%.**

3.3.5 VAL2 - CMR Qflow

1-Proportion test results: P-value=0.000
P-Value< 0.05 therefore, the Null hypothesis can be rejected. This means that the criterion was met: **at least 80% of the sample was ranked as Clinically Acceptable in a confidence level of 95%.**

3.3.6 VAL2 - CMR Spatial

During the validation period a total of 65 evaluation forms were collected. All studies contained T1w and T2w multi-slice, single-phase short axis MR images
Out of 65 forms, 63 were graded as clinically acceptable (Perfect, Good and Acceptable) and 2 were graded as clinically unacceptable (Bad).

The following parameters were the input to the 1-Proportion test (using Minitab 19):

Null hypothesis = there are less than 80% case forms ranked as clinically acceptable.
Reported events = 63
Sample size =65
Confidence level = 95%

Results analysis:
Defects were open in Azure DevOps (TFS) for dissatisfied scores for clinically unacceptable Refer to section defects of this document.

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Test and CI for One Proportion

Method

p: event proportion
Exact method is used for this analysis.

Descriptive Statistics

N	Event	Sample p	95% Lower Bound for p
65	63	0.969231	0.906293

Test

Null hypothesis	H_0: p = 0.8
Alternative hypothesis	H_1: p > 0.8

P-Value
0.000

1-Proportion test results: P-value=0.000
P-Value< 0.05 therefore, the Null hypothesis can be rejected. This means that the criterion was met: **at least 80% of the sample was ranked as Clinically Acceptable in a confidence level of 95%.**

3.3.7 VAL2 - CMR Functional Long Axis

During the validation period a total of 65 evaluation forms were collected. All studies containing single and multi-slice functional Long Axis cardiac cine data.
All 65 forms were graded as clinically acceptable (Perfect, Good and Acceptable).

The following parameters were the input to the 1-Proportion test (using Minitab 19):
Null hypothesis = there are less than 80% case forms ranked as clinically acceptable.
Reported events = 65
Sample size =65
Confidence level = 95%

Results analysis:
No Defects from validation

Test and CI for One Proportion

Method

p: event proportion
Exact method is used for this analysis.

Descriptive Statistics

N	Event	Sample p	95% Lower Bound for p
65	65	1.000000	0.954958

Test

Null hypothesis	H_0: p = 0.8
Alternative hypothesis	H_1: p > 0.8

P-Value
0.000

1-Proportion test results: P-value=0.000

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P-Value< 0.05 therefore, the Null hypothesis can be rejected. This means that the criterion was met: **at least 80% of the sample was ranked as Clinically Acceptable in a confidence level of 95%.**

3.3.8 VAL3 - CMR Functional LV RV

During the validation period a total of 65 evaluation forms were collected. All studies containing multi-slice short axis and axial cardiac MR cine data
All 65 forms were graded as clinically acceptable (Perfect, Good and Acceptable).

The following parameters were the input to the 1-Proportion test (using Minitab 19):

Null hypothesis = there are less than 80% case forms ranked as clinically acceptable.
Reported events = 65
Sample size =65
Confidence level = 95%

Results analysis:
No Defects from validation

Test and CI for One Proportion

Method

p: event proportion
Exact method is used for this analysis.

Descriptive Statistics

N	Event	Sample p	95% Lower Bound for p
65	65	1.000000	0.954958

Test

Null hypothesis	H_0: p = 0.8
Alternative hypothesis	H_1: p > 0.8

P-Value
0.000

1-Proportion test results: P-value=0.000
P-Value< 0.05 therefore, the Null hypothesis can be rejected. This means that the criterion was met: **at least 80% of the sample was ranked as Clinically Acceptable in a confidence level of 95%.**

3.3.9 VAL3 - CMR Mapping

During the validation period a total of 65 evaluation forms were collected. All studies containing MR T1 native, T1 Enhanced, T2 and T2* MR data.
All 65 forms were graded as clinically acceptable (Perfect, Good and Acceptable).

The following parameters were the input to the 1-Proportion test (using the Minitab 19):

Null hypothesis = there are less than 80% case forms ranked as clinically acceptable.
Reported events = 65

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Sample size =65
Confidence level = 95%

Results analysis:
No Defects from validation

Test and CI for One Proportion

Method

p: event proportion
Exact method is used for this analysis.

Descriptive Statistics

N	Event	Sample p	95% Lower Bound for p
65	65	1.000000	0.954958

Test

Null hypothesis	H_0: p = 0.8
Alternative hypothesis	H_1: p > 0.8

P-Value
0.000

1-Proportion test results: P-value=0.000
P-Value< 0.05 therefore, the Null hypothesis can be rejected. This means that the criterion was met: **at least 80% of the sample was ranked as Clinically Acceptable in a confidence level of 95%.**

3.3.10 VAL3 - CMR Temporal

During the validation period a total of 65 evaluation forms were collected. All studies containing single and multi-slice Temporal enhancement (rest or stress acquisitions and/or rest-stress comparison of multi-slice dynamic cardiac MR acquisition data.
All 65 forms were graded as clinically acceptable (Perfect, Good and Acceptable).

The following parameters were the input to the 1-Proportion test (using Minitab 19):

Null hypothesis = there are less than 80% case forms ranked as clinically acceptable.
Reported events = 65
Sample size =65
Confidence level = 95%

Results analysis:
No Defects from validation

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Test and CI for One Proportion

Method

p: event proportion
Exact method is used for this analysis.

Descriptive Statistics

N	Event	Sample p	95% Lower Bound for p
65	65	1.000000	0.954958

Test

Null hypothesis	H_0: p = 0.8
Alternative hypothesis	H_1: p > 0.8

P-Value
0.000

1-Proportion test results: P-value=0.000
P-Value< 0.05 therefore, the Null hypothesis can be rejected. This means that the criterion was met: **at least 80% of the sample was ranked as Clinically Acceptable in a confidence level of 95%.**

3.3.11 VAL3 - CT Calcium Automatic Analysis (Auto CS)

During the validation period a total of three reports of CT Calcium Automatic Analysis (preliminary named Auto CS) were collected.
Each evaluator reviewed one case of patient who underwent CT chest scan with incidental finding of calcium.
Each evaluator graded a single form according to his impression from the application.
All steps are marked as Pass.

No comment was made by the evaluators.

Results analysis: Pass
Dissatisfied score: None
Add a description of defect/RFC: None

3.3.12 VAL3 – Spectral AVA

During the validation period a total of three reports of Spectral AVA were collected.
Each evaluator reviewed one Spectral CTA case with pathology.
Each evaluator graded a single form according to his impression from the application.
All steps are marked as Pass.

No comment was made by the evaluators.

Results analysis: Pass
Dissatisfied score: None
Add a description of defect/RFC: None

3.3.13 VAL3 – MR Longitudinal Brain Analysis (MR CVS)

During the validation period a total of three reports of MR Longitudinal Brain Analysis (CVS)were collected.
Each evaluator reviewed one case with registered scans of 3D Flair and 3Dswi-epi sequence of MS adult
ach evaluator graded a single form according to his impression from the application.

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All steps are marked as Pass.

No comment was made by the evaluators.

Results analysis: Pass
Dissatisfied score: None
Add a description of defect/RFC: None

3.4 Usability Engineering – Results

Refer to DHF380228 ISP13 Usability Engineering Plan

4 Defects

All defects assigned for clinically unacceptable ranking are related to functionality issues or evaluator dissatisfaction. None was found to be hazard or safety related issues.
No defects were opened for any VAL3 validation forms.

ISP13 defect from validation forms

5 Conclusion

The ISP13 applicable applications was validated internally and externally by clinical evaluators as defined in section "Design Validation Top Level Objectives& Statistical Measurement" of the validation plan [DHF380226] Results for requirements that were identified in the validation Plan [DHF380226] section traceability to URS as items that can be tested according to the Verification process can be found in the Verification Reports [D000889689]

VAL1:

Server	HX configurations at stated in the PRS [DHF380211]
SW	13.0.1.11183

VAL2:

Server	HX , IX and demo laptop configurations at stated in the PRS [DHF380211]
SW	13.0.2.11236

VAL3:

Server	HX , IX and demo laptop configurations at stated in the PRS [DHF380211]
SW	13.0.3.11559

Based on the completion of design validation objectives described in validation plan, we have completed the validation of:

1. CT ASPECTS (VAL1)
2. CT Liver (VAL2)

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3. CT CCA (VAL2)
4. CMR Viewer (VAL2)
5. CMR Qflow (VAL2)
6. CMR Spatial (VAL2)
7. CMR Functional Long Axis (VAL2)
8. CMR Functional LV RV (VAL3)
9. CMR Mapping (VAL3)
10. CMR Temporal (VAL3)
11. CT CAA (CT Auto. CS VAL3)
12. Spectral AVA (VAL3)
13. MR Longitudinal Brain Analysis (VAL3)

provided evidence that the intended uses and defined user needs were met.
It has been concluded that above list of ISP13 applications were validated as described in DHF380226 ISP13 validation plan.
There are no major changes in workflow or functionally of the validated applications which will require re-validated from in RFD.

Based on the completion of Validation objectives described in the ISP13 Validation Plan the CT ASPECTS, CT Liver, CT CCA , CMR View, CMR Qflow and CMR Spatial CMR functional Long Axis , CMR Functional LV RV, CMR Mapping, CMR Temporal, CS CAA, Spectral AVA, MR Longitudinal Brain Analysis CVS applications are ready for release .

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6 Record History

Rev	Release Date	Author	Description of changes
A	In Windchill	Hagit Davidesco	Post VAL1 validation for CT ASPECTS Score
B	In Windchill	Hagit Davidesco	Post VAL2 validation for CT Liver, CT CCA , CMR Viewer, CMR Qflow, CMR Functional Long Axis and CMR Spatial
C	In Windchill	Hagit Davidesco	Post VAL3 validation for CT CAA, Spectral AVA , MR Longitudinal Brain Analysis , CMR Functional LV RV, CMR Mapping and CMR Temporal

7 Record Review and Approval

Name Author	Function Author	Review Date *DD-MMM-YYYY*	Signature
Hagit Davidesco	Validation Lead	In Windchill	In Windchill

Name Reviewer(s)	Function Reviewer(s)	Review Date(s) *DD-MMM-YYYY*	Signature(s)
Shipra Shivangi	Quality Engineer	In Windchill	In Windchill
Efrat Kliper Goldin	Product Owner	In Windchill	In Windchill
Keren Afek	Regulatory Affairs Specialist	In Windchill	In Windchill
Sujesh Paravatty	Clinical Lead	In Windchill	In Windchill
Emilie Santos	Validation Manager	In Windchill	In Windchill

Name Approver(s)	Function Approver(s)	Approval Date(s) DD-MMM-YYYY	Signature(s)
Alina Shapira	Director of ICAP Product Management	In Windchill	In Windchill
Eyal Biran	Program Manager	In Windchill	In Windchill

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8 Annexes / Appendices

8.1 Appendix A: Validation Summary of Evidence/ Results

A.1 **VAL1 : Actual Number of Studies Performed per Site and Validation Phase**

Sites	System	Total
Internal	HX	3
Total		3

VAL2 : Actual Number of Studies Performed per Site and Validation Phase

Sites	System	Total
Internal	HX- Demo Laptop	118
Sheba	IX	15
NH	HX- Demo Laptop	108
Rosario	HX	15
Total		**256**

VAL3 : Actual Number of Studies Performed per Site and Validation Phase

Sites	System	Total
Internal	HX- Demo Laptop	94
Sheba	IX	12
NH	HX- Demo Laptop	65
Rosario	HX	20
Total		**191**

A.2 Actual Number of Studies Performed per Exam Type

EXAM TYPE	Internal
CT ASPECTS	3

VAL2:

EXAM TYPE	Internal	Sheba	NH	Rosario	Cologne	Total	Total NO. Of evaluators
CMR Viewer	30	3	26	2	4	**65**	**14**
CMR Qflow	15	12	26	5	7	**65**	**17**
CMR Functional Long Axis	27	2	26	3	7	**65**	**15**
CMR Spatial	22	6	26	5	6	**65**	**17**
CT Liver	0	1	2	0	0	**3**	**3**
CT CCA	0	1	2	0	0	**3**	**3**

VAL3:

EXAM TYPE	Internal	Sheba	NH	Rosario	Total	Total NO. Of evaluators
CMR Functional LV RV	35	4	18	8	**65**	**13**
CMR Mapping	35	4	18	8	**65**	**13**
CMR Temporal	24	4	33	4	**65**	**14**
CT CAA	0	0	2	1	**3**	**3**

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Spectral AVA	1	0		2	**3**	**3**
MR Longitudinal Brain Analysis CVS	1	0	2		**3**	**3**

A.3 Actual Number of Studies Performed per Feature/Application

Site	VAL1	VAL2	VAL3
Internal	3	94	96
Sheba	0	25	12
NH	0	108	73
Rosario	0	15	23
Cologne	0	24	0
Total	**3**	**266**	**204**

A.4 The results of the primary and secondary objectives

User Questionnaire

VAL1 CT ASPECTS - 3 users PASS
VAL2 – CT Liver- 3 users PASS
VAL2 – CT CCA - 3 users PASS
VAL3 – CT Auto CS- 3 users PASS
VAL3 – Spectral AVA - 3 users PASS
VAL3 – MR CVS- 3 users PASS

Statistical Validation

Required number of forms per application is 65
Ranking Scale in the applicable sections below:

Clinically Acceptable	++ = perfect
	+ = Good
	0 = Acceptable
Clinically Unacceptable	- = Bad
	-- = Unacceptable

VAL2:

Site Name		CMR Viewer				
	++ Perfect	+ Good	0 Acceptable	- Bad	-- Unacceptable	Total No. of Forms
Internal	19	11	0	0	0	30
Sheba	0	1	2	0	0	3
NH	2	23	1	0	0	26
Rosario	2	0	0	0	0	2
Cologne	0	4	0	0	0	4
Total	23	39	3	0	0	**65**

Site Name		CMR Qflow	

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	++ Perfect	+ Good	0 Acceptable	- Bad	-- Unacceptable	Total No. of Forms
Internal	8	6	1	0	0	15
Sheba	10	2	0	0	0	12
NH	7	17	2	0	0	26
Rosario	4	1	0	0	0	5
Cologne	2	4	1	0	0	7
Total	31	30	4	0	0	**65**

Site Name		**CMR Functional Long Axis**				
	++ Perfect	+ Good	0 Acceptable	- Bad	-- Unacceptable	Total No. of Forms
Internal	18	9	0	0	0	27
Sheba	0	0	2	0	0	2
NH	2	23	1	0	0	26
Rosario	3	0	0	0	0	3
Cologne	1	5	1	0	0	7
Total	24	37	4	0	0	**65**

Site Name		**CMR Spatial**				
	++ Perfect	+ Good	0 Acceptable	- Bad	-- Unacceptable	Total No. of Forms
Internal	6	7	7	2	0	22
Sheba	0	3	3	0	0	6
NH	2	21	3	0	0	26
Rosario	1	3	1	0	0	5
Cologne	1	3	2	0	0	6
Total	10	37	16	2	0	**65**

VAL3:

Site Name		**CMR Functional LV RV**				
	++ Perfect	+ Good	0 Acceptable	- Bad	-- Unacceptable	Total No. of Forms
Internal	21	12	2	0	0	35
Sheba	4	0	0	0	0	4
NH	11	7	0	0	0	18
Rosario	3	5	0	0	0	8
Total	39	24	2	0	0	**65**

Site Name		**CMR Mapping**				
	++ Perfect	+ Good	0 Acceptable	- Bad	-- Unacceptable	Total No. of Forms

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Internal	12	19	4	0	0	35
Sheba	0	4	0	0	0	4
NH	0	16	2	0	0	18
Rosario	6	2	0	0	0	8
Total	18	41	6	0	0	**65**

Site Name		**CMR Temporal**				
	++ Perfect	+ Good	0 Acceptable	- Bad	-- Unacceptable	Total No. of Forms
Internal	9	12	3	0	0	24
Sheba	0	4	0	0	0	4
NH	1	31	1	0	0	33
Rosario	2	2	0	0	0	4
Total	12	49	4	0	0	**65**

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8.2 Validation Forms

All issues from evaluation forms were analyzed and assigned a defect/RFC if applicable, to be handled by the defect management process in Azure DevOps (TFS). Refer to section defects in this document

VAL1	
CT ASPECTS Score	ISP13 CT ASPECTS Validaiton forms.pd

VAL2	
CMR – Viewer, Qflow, Spatial, Functional Long Axis	ISP13 VAL2 CMR Cologne.pdf ISP13 VAL2 CMR DHZB.pdf ISP13 VAL2 CMR internal.pdf ISP13 VAL2 CMR NH.pdf ISP13 VAL2 CMR Sheba.pdf ISP13 VAL2 CMR Lyon.pdf ISP13 VAL2 CMR Rosario.pdf
CT Liver	ISP13 CT Liver validaion forms.pdf
CT CCA	ISP13 CCA valiation forms.pdf

VAL3	
CMR – Functional LV RV, Mapping and Temporal	ISP13 VAL3 CMR_Berlin.pdf ISP13 VAL3 CMR_Israel.pdf ISP13 VAL3 CMR_Lyon.pdf ISP13 VAL3 CMR_NH.pdf ISP13 VAL3 CMR_Rosario.pdf ISP13 VAL3 CMR_USA.pdf
CT Auto CS	ISP13 VAL3 CT CAA forms.pdf

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Spectral AVA	ISP13 VAL3 sAVA forms.pdf
MR	ISP13 VAL3 CVS.pdf

8.3 Training Forms

SL. No	Comments/ Action Item	Responsibility	Status	Due date (DD-MMM-YYYY)	Actual Closure date (DD-MMM-YYYY)	Closure Remarks
1.	page 2 – missing trainer from Philips signature	validation lead	Approved	Rev B	N/A	N/A

VAL1 – CT ASPECTS Score	**VAL2 CMR – Viewer, Qflow, Spatial Functional Long Axis**	**VAL2 – CT Liver**	**VAL2 – CT CCA**
ASPECT training forms.pdf	ISP13 VAL2 CMR training forms.pdf	ISP13 CT Liver training forms.pdf	ISP13 CCA training forms.pdf
VAL3 CMR – Functional LV RV, Mapping and Temporal	**VAL3 – CT Auto CS**	**VAL3 – Spectral AVA**	**VAL3 – MR Longitudinal Brain Analysis (MR CVS)**
ISP13 VAL3 CMR training forms.pdf	ISP13 Val3 Auto CS training.pdf	ISP13 Val3 sAVA training.pdf	ISP13 VAL3 CVS training forms.pdf

8.4 Validation Site Satisfaction Survey

NA, there were only 2 test serves installed at the 2 hospitals: Sheba and Rosario.
As these systems are labeled and marked "Not for Clinical Use" there were used only for validation purposes, collecting feedback and validation forms, while Philips representative were visiting.

End of Document

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